Real-time Communication Security-IPSec: Authentication Header, Encapsulating Security Payload Protocols

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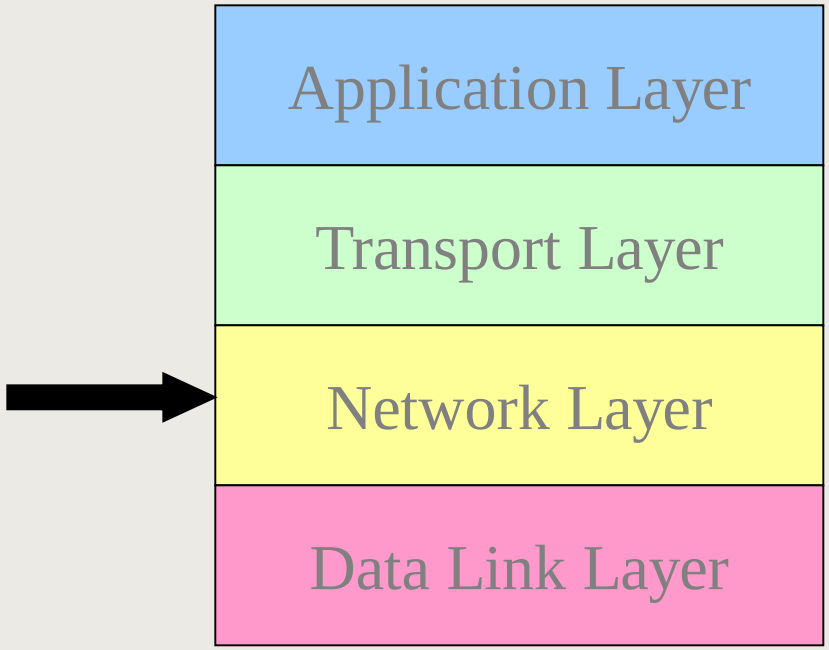
TCP/IP Protocol Stack



For detail explanation:
https://www.youtube.com/watch?v=ScxCFzxVel8

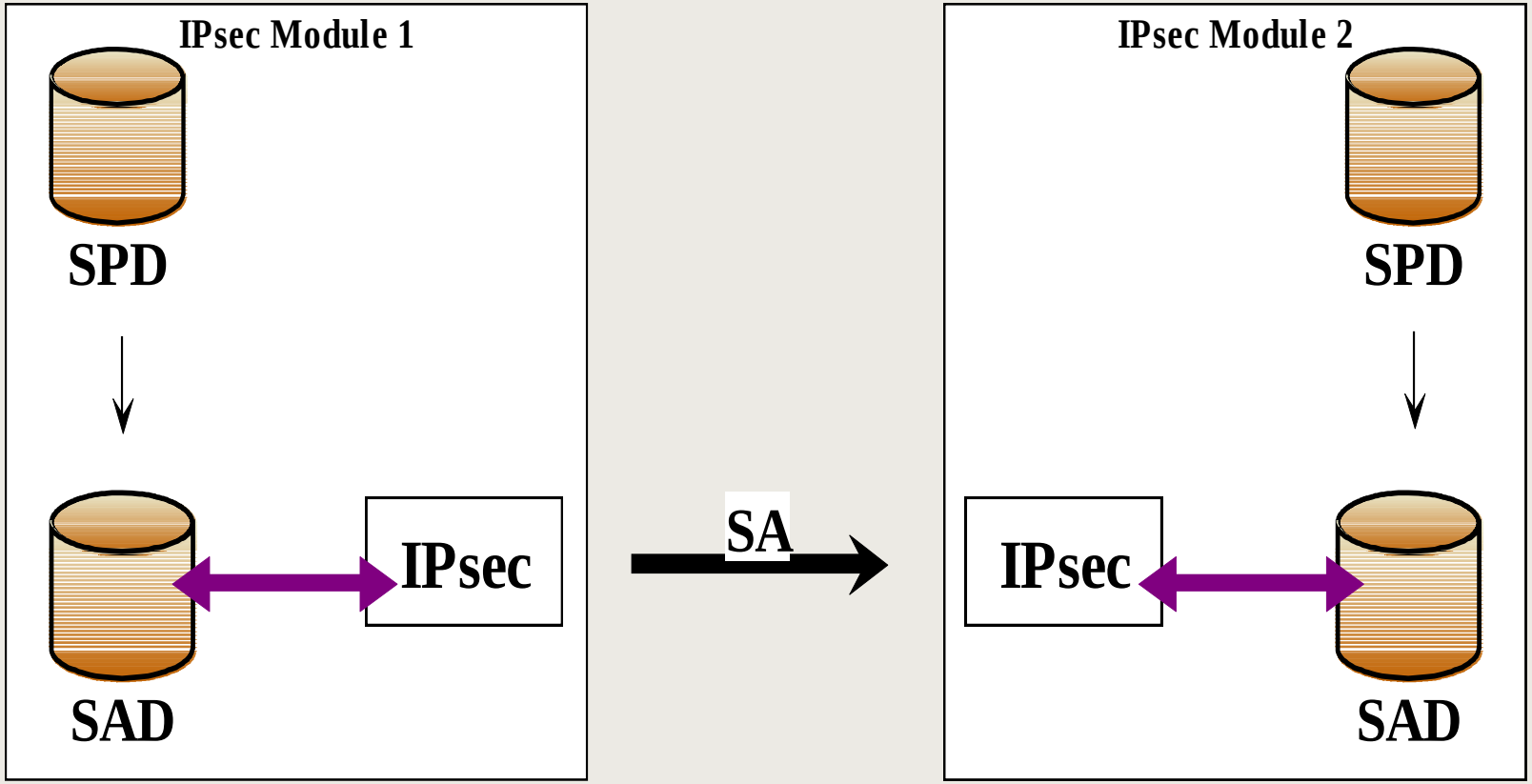
IPSec Architecture

- Set of security services offered by IPSec include
  - Connectionless integrity
  - Data origin authentication
  - Protection against replay attacks
  - Confidentiality
  - Limited traffic flow confidentiality
- The services can be used alone or in combination
- Security is provided for protection of the IP and/or upper layer protocols(tcp, udp)
- IPSec can be thought of as a software or hardware module that is implemented in either a host or a security gateway (router or firewall)

IPSec Architecture

- IPSec module is used to manage security for individual connections to other modules
  - Security Policy Database (SPD) provides specifications of the security services to be applied to each packet
  - Security Association Database (SAD) contains the security parameters (encryption algorithms, mode used, initialization data, session keys) used to enforce a specific policy
  - A connection from one module to another is created through a security association (SA) that corresponds to an entry in the SAD
  - An SA is a uni-directional connection that defines the type of security services and mechanisms used between two modules

IPSec Architecture



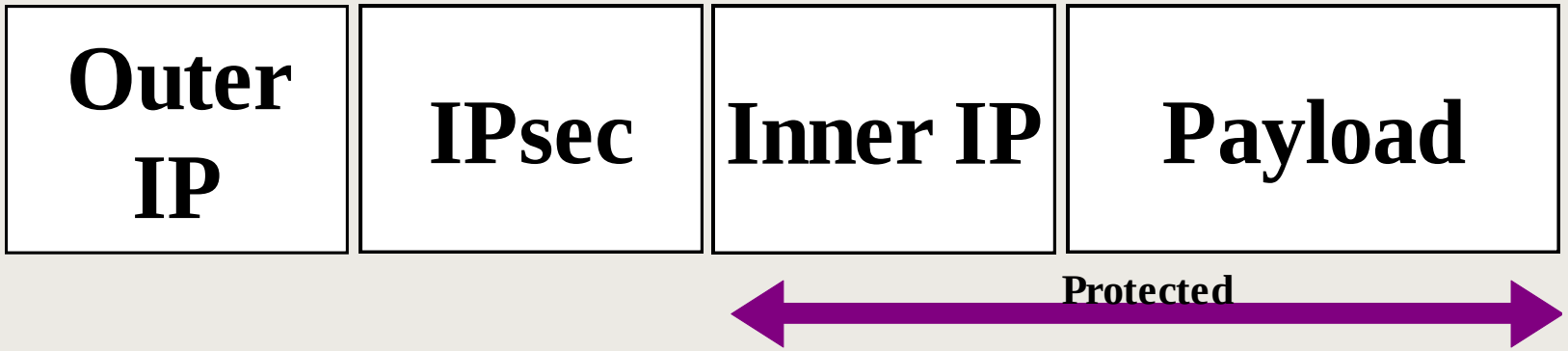
IPSec Protocols

- The protocols used to provide security are the **Authentication Header (AH)** and **Encapsulating Security Payload (ESP)**
- Each protocol can be used in one of two modes
  - Transport mode – used to protect upper layer payloads of an IP packet (tcp, udp)
  - Tunnel mode – used to protect an entire IP packet including its payload (VPN)
- Transport mode is used as an SA between two hosts
- Tunnel mode is used as an SA between two gateways or a host and gateway

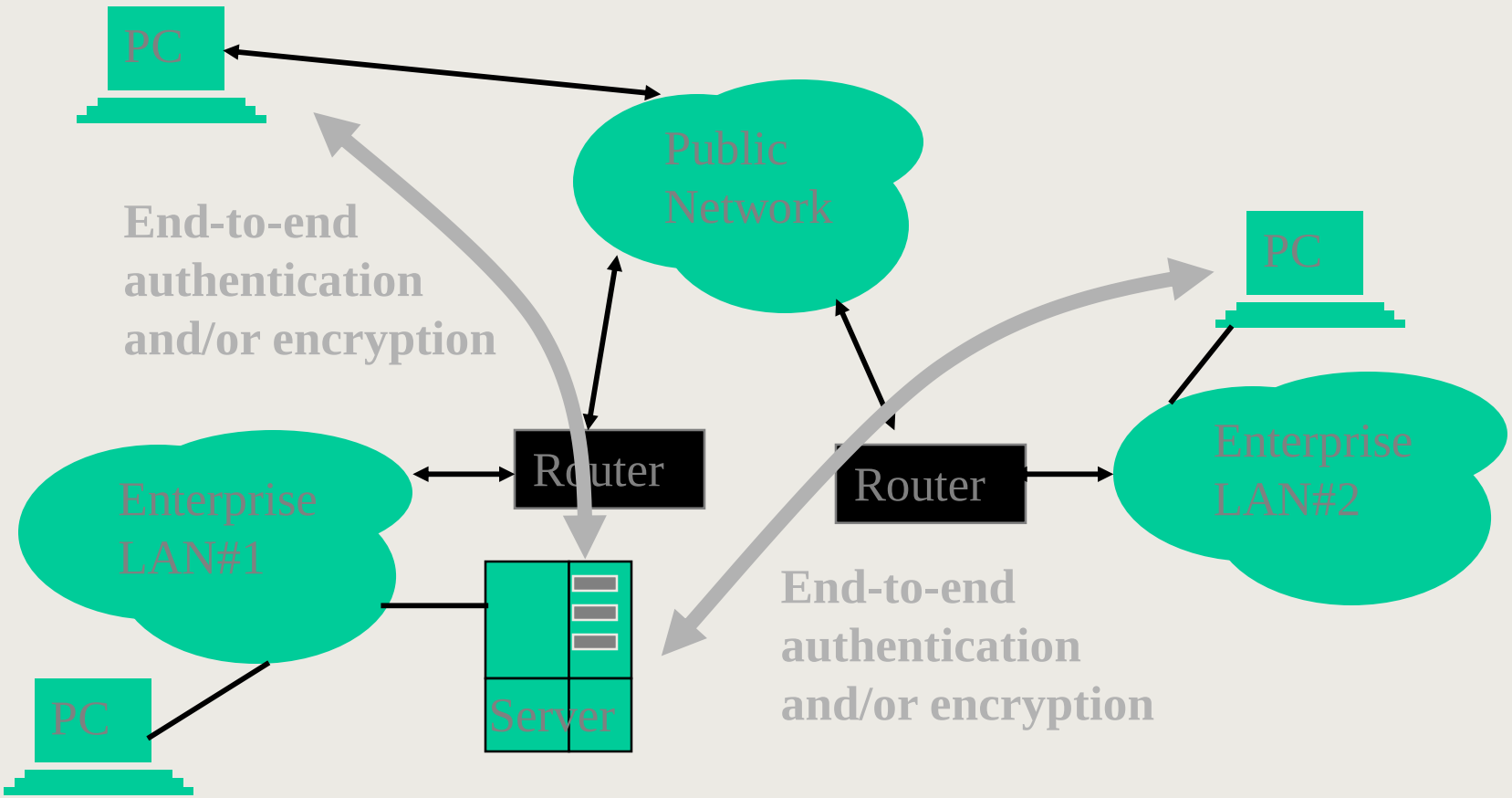
IPSec Protocols

- Transport Mode (upper level protocols)

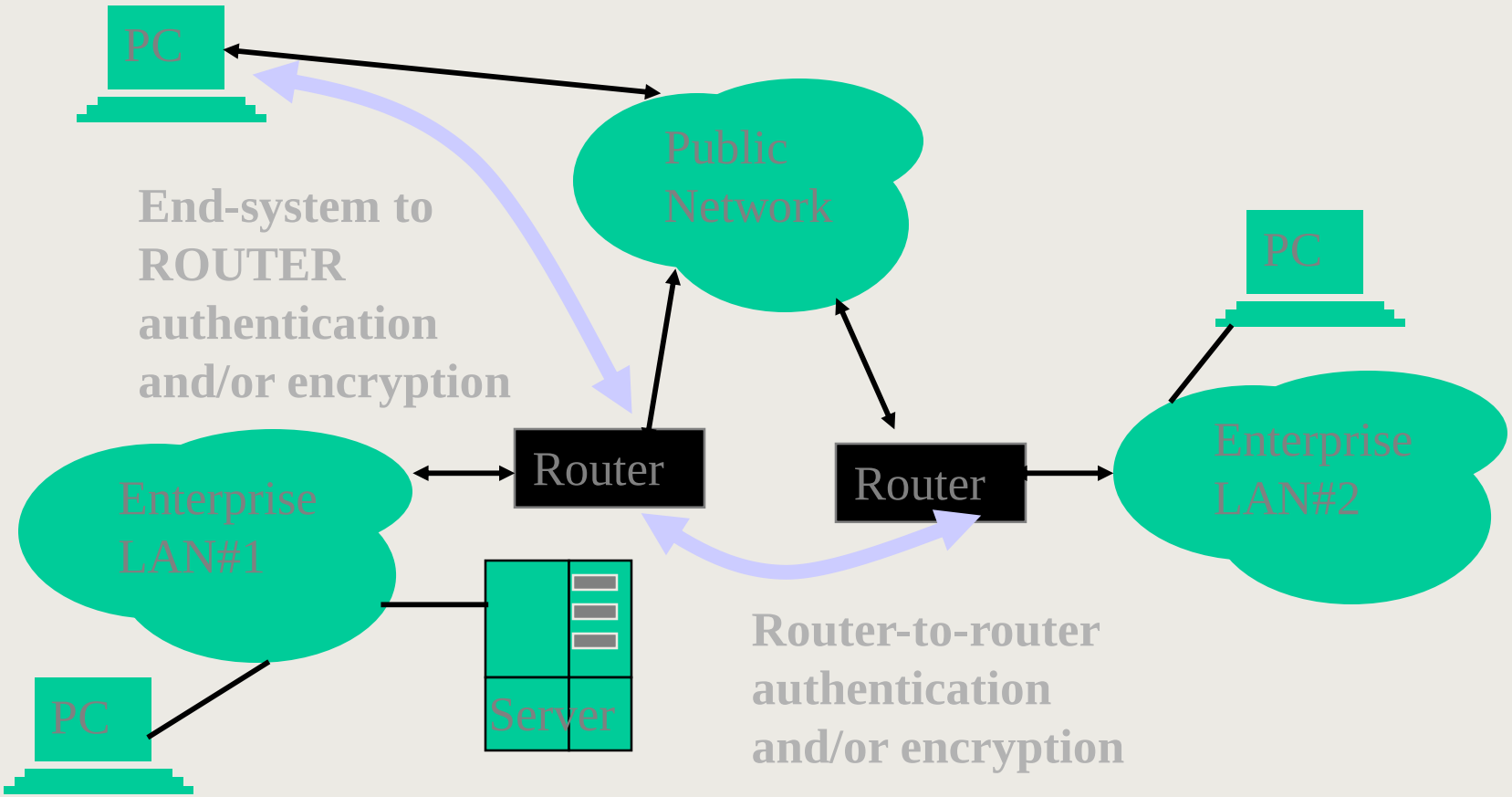
- Tunnel Mode (entire IP packet)



Transport Mode IPSec



Tunnel Mode IPSec



IPSec Protocols

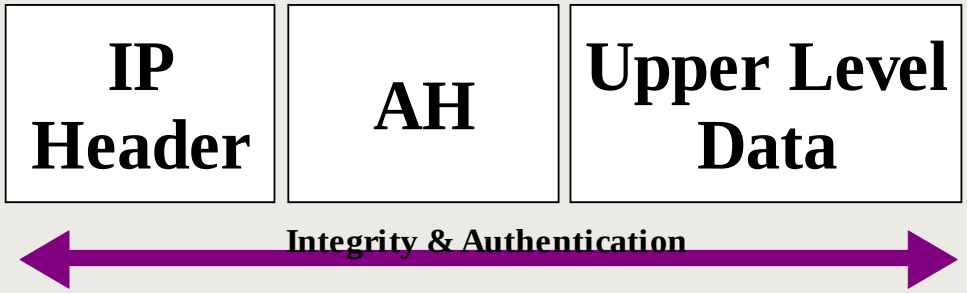
- AH is used to provide
  - Connectionless integrity and data origin authentication (integrity)
  - Optional anti-replay service
- ESP is used to provide
  - Confidentiality and (integrity) connectionless integrity and data origin authentication
  - Connectionless integrity and data origin authentication (integrity)
  - Limited traffic flow confidentiality
  - Optional anti-replay service

IPSec Protocols

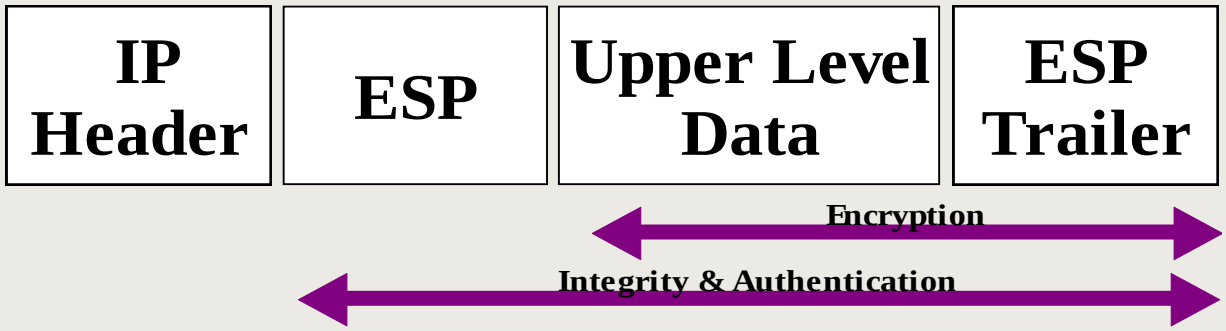
- Integrity Algorithm (AH, ESP)
  - Hashed Message Authentication Code (160 bit key)
- Confidentiality Algorithm (ESP)
  - AES CBC mode (128 bit key – 256 bit key)
- **Transport Mode Protection**
- **AH - Integrity**
  - Immutable sections of the IP header, the AH header, and the upper level data
- **ESP - Integrity**
  - The ESP header, the upper level data, and the ESP trailer
- **ESP – Confidentiality**
  - The upper level data, and the ESP trailer

IPSec Protocols

- Transport Mode (AH)



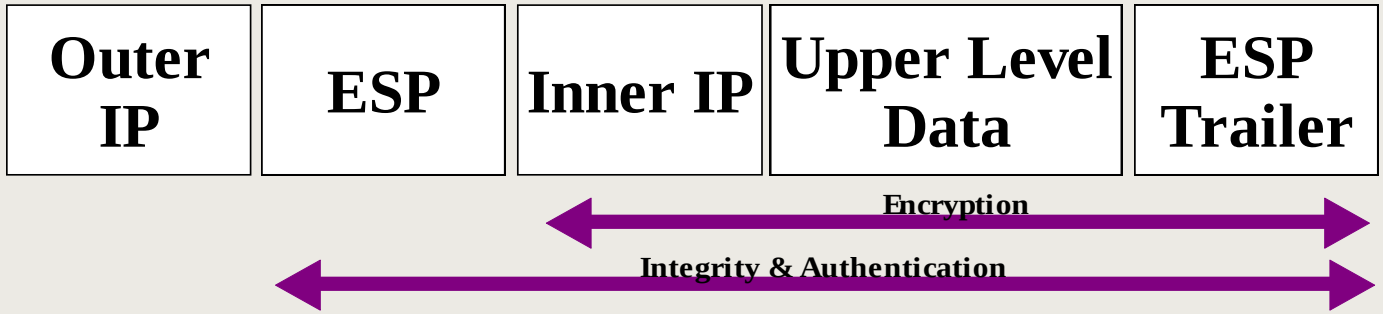
- Transport Mode (ESP)



IPSec Protocols

- Tunnel Mode Protection
- AH - Integrity
  - Immutable sections of the outer IP header, the AH header, and the entire inner IP packet
- ESP - Integrity
  - The ESP header, the entire inner IP packet, and the ESP trailer
- ESP – Confidentiality
  - The entire inner IP packet, and the ESP trailer

IPSec Protocols

- Tunnel Mode (AH)

- Tunnel Mode (ESP)